Oct 26, 2026

Virtual Gold (Hybrid Architecture) x CMU Capstone: Weekly Sync

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Attachments Virtual Gold (Hybrid Architecture) x CMU Capstone: Weekly Sync client-update-2026-10-26.pdf tier-table-client-review-en.docx

Meeting records Transcript Recording

Summary

Rules progress reviewed and model evaluations presented via API updates and timeline discussions

Rules and Tier Review
Rules performed well on pattern-based data and tier table proposed with commercial categories.

Model Evaluation Comparison
Qwen combined with Gliner reduced private text leakage significantly while accurately capturing confidential facts.

API and Timeline Integration
Server-side API updates presented with AES encryption and integration planned for next week.

Decisions

Aligned

- **Tier table severity categories approval** The participants agreed to approve the tier table representing data severity categories to proceed with development.
- **API request ID and hash key workflow** The group agreed that the redaction service will generate and return a hash key for the client to use as a unique conversation and request identifier during message restoration.
- **System integration and testing timeline** The participants agreed to target system integration by next Monday to begin testing.
- **Encryption standard for redaction restoration** The team agreed to use encryption and decryption with a single hash key per request for redaction restoration moving forward.

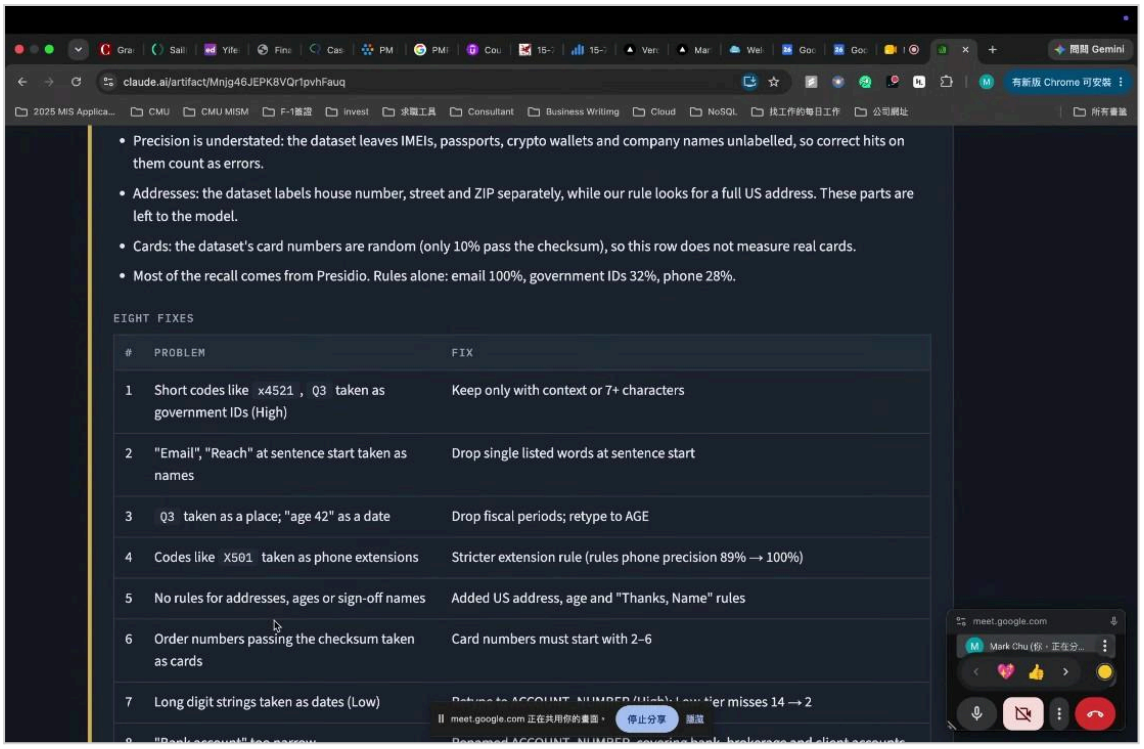
Next steps

- ☐ [Mark Chu] Update Timeline: Revise the project timeline based on recent progress. Share the updated schedule on Slack to ensure team alignment.
- ☐ [Anmol Sahu] Send Flow Diagram: Draw a flow diagram illustrating the API and hash key process. Email the document to the team for review.
- ☐ [Mark Chu] Share Meeting Summary: Consolidate meeting points and decisions. Distribute the summary on Slack for team synchronization.
- ☐ [Karina Yang] Update Test Categories: Update the test set categories following the confirmation of the tier table. Score the service on the 60 development emails.
- ☐ [Anmol Sahu] Evaluate New Models: Research and evaluate additional models including Gemma 4 and Mistral. Perform these tests once the computing credits are secured.
- ☐ [The group] {Integrate system}: Combine system components by Monday of the coming week.

- [ ] [The group] {Start testing}: Initiate assessment of the integrated system.
- [ ] [The group] {Request resources}: Contact Carnegie Mellon University to obtain hardware necessary for the system.
- [ ] [The group] {Resolve issues}: Coordinate on Slack to identify and fix any problems related to system resources.
- [ ] [Anmol Sahu] {Email contract}: Provide the necessary contract documentation to Aarvin George.

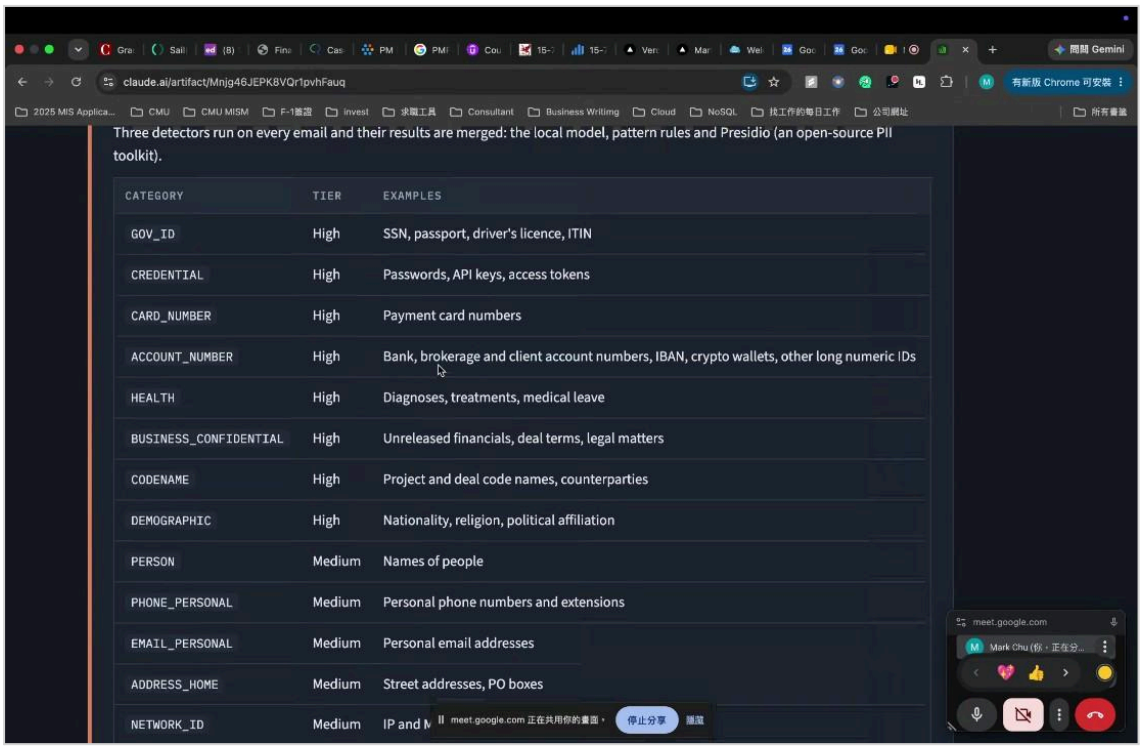
Details

Rules and Procedures Update: Yudi discussed progress on rules and procedures tested on a private dataset, specifically the AI for privacy API masking dataset (00:02:33). Yudi reported that the rules performed well on pattern-based data like emails, but struggled with simulated addresses and real call numbers (00:04:13). Based on these results, Yudi fixed issues such as short codes like Q3 being incorrectly classified as government IDs, updating the requirement to mandate more than seven characters (00:05:47).



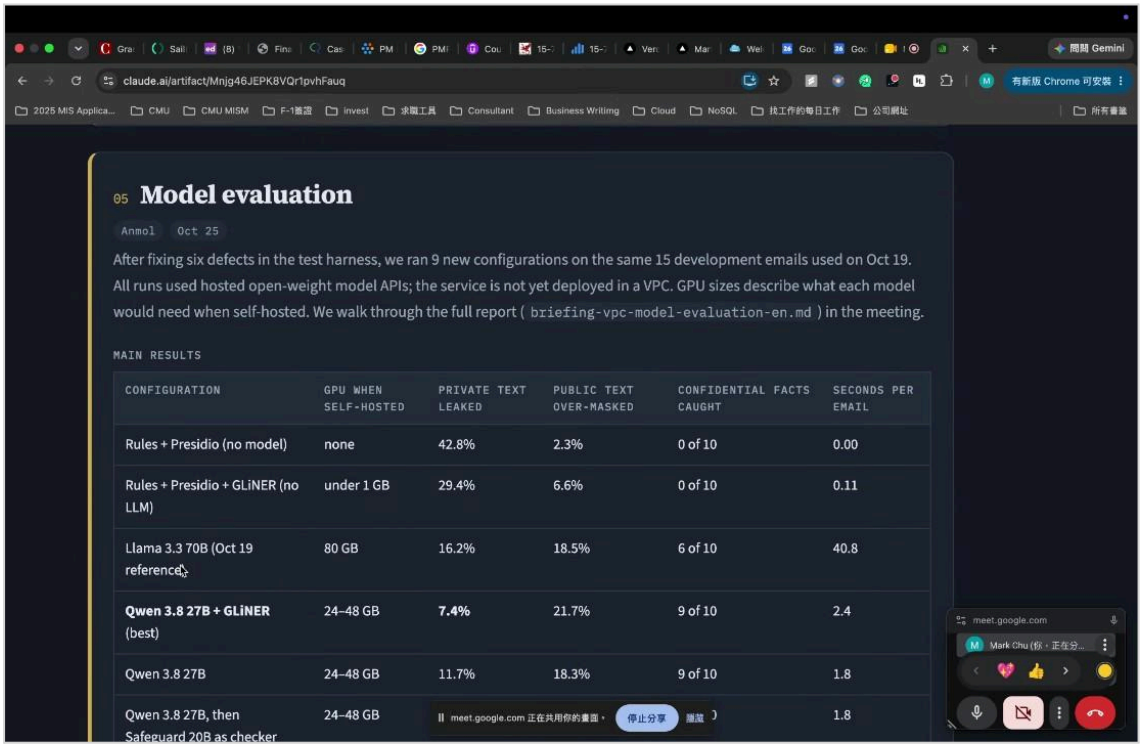
Tier Table Review and Approval: Yudi proposed a tier table that serves as the

foundation for their classification work (00:05:47). Aarvin George reviewed the document and stated that it looked good enough, represented severity correctly, and presented no blockers or objections to proceeding (00:07:27). Mark Chu noted that commercial information in the tier table was weak and invited team members on Slack to add categories for commercial data like account numbers, organizations, and amounts (00:08:31).

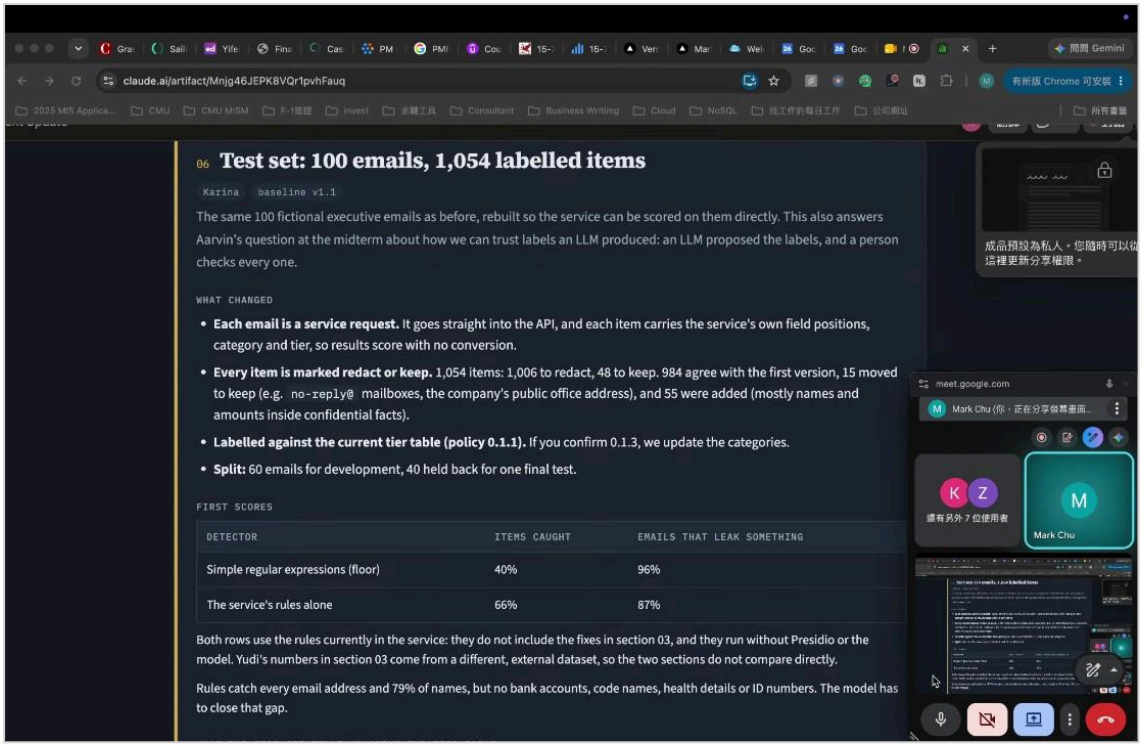


Model Evaluation and Performance Comparison: Anmol Sahu presented evaluations for various named entity recognition and redaction models, noting that rules plus Presidio leaked 42.8% of private text. Adding the Gliner inference model, which requires under 1 GB of GPU, significantly reduced private text leakage (00:10:02). Anmol evaluated Llama 3.3 (70 billion parameters), which leaked 16.2% of private text with high latency. The best-performing configuration was Qwen (3.827 billion parameters) combined with Gliner, which requires 24 to 48 GB of GPU when self-hosted, reduced private text leakage to 7.4% while redacting over 92% of private text, and accurately captured 9 out of 10 confidential facts (00:11:50) (00:22:52). Inderpal Bhandari asked about missed facts, and Anmol explained that a medical diagnosis sentence was missed because it was categorized as health instead of fact and due to a strict coverage threshold (00:13:33) (00:18:19). Anmol also tested Safeguard 20B and GPT-OSS 120B, and noted that they requested credits from CMU computing services to evaluate additional models like Gemma 4 and Mistral (00:22:52). Aarvin confirmed that they would use the chosen local

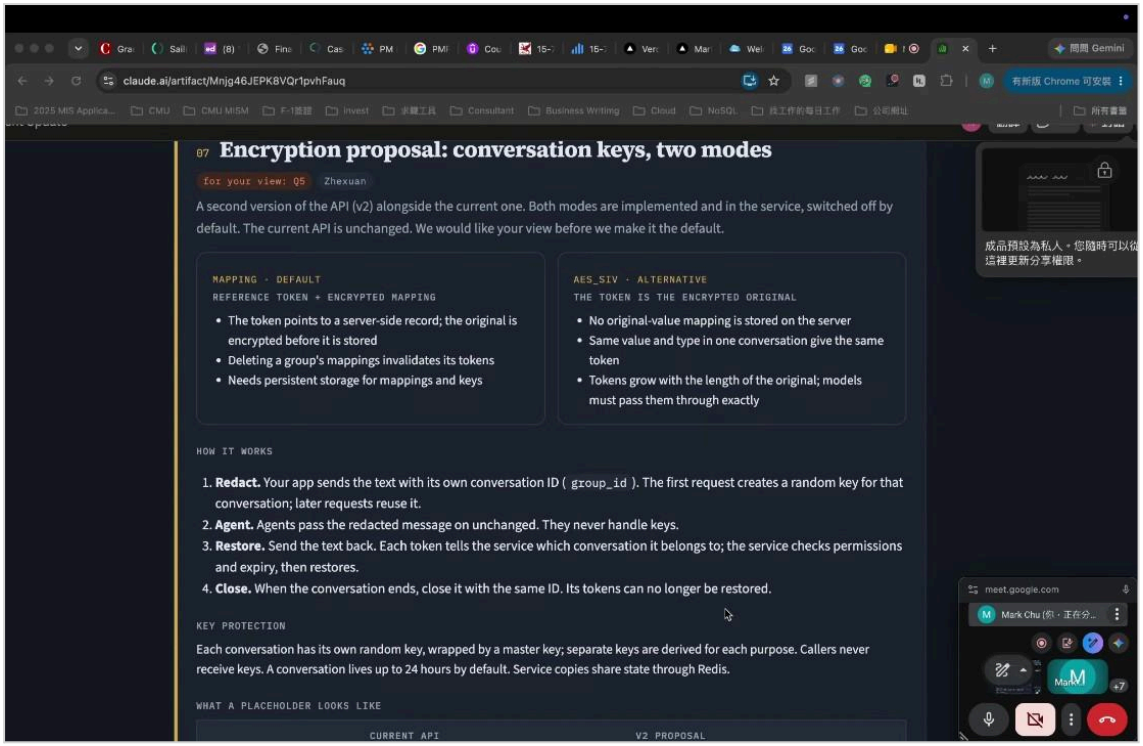
model without needing runtime switches (00:25:59).



Test Set Rebuilding and Validation: Karina Yang explained that they rebuilt a 100-email test set of fictional executive emails containing 155 items in total, with 107 items designated for redaction and 48 items to remain visible, addressing questions about label trust (00:27:09). Karina split the dataset into 60 development emails and 40 held-back emails for final testing (00:28:32). Karina reported that basic detectors caught 40% of items, while Yudi's current rules caught 66%. Karina noted that 86 low-tier items, including 67 company names, 12 cities, 6 dates, and 1 age, would be redacted under current rules and potentially reach the cloud (00:30:17). Karina stated they will update test set categories once the tier table is confirmed and score the service on development emails (00:31:56).

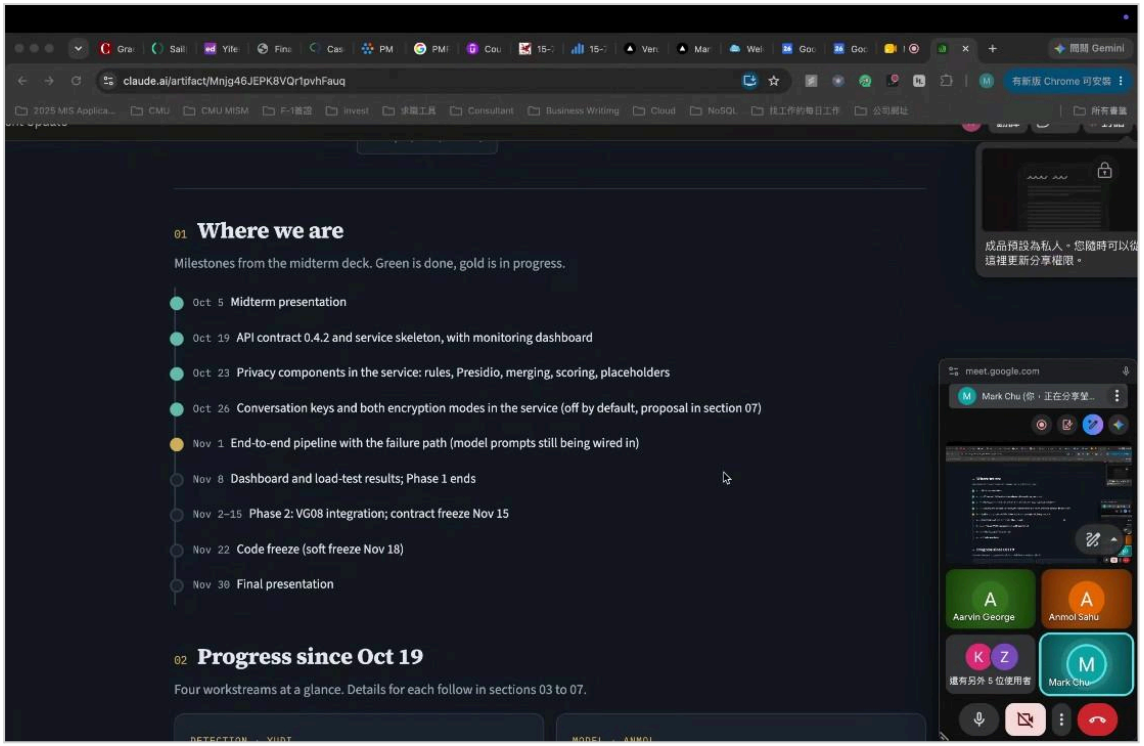


API Workflow and Encryption Proposal: Mark Chu presented server-side API updates, explaining that they removed previous redaction IDs and shifted to a workflow where the app sends text with a conversation identifier (00:33:42). Mark proposed using Advanced Encryption Standard (AES) encryption with the conversation ID as a passphrase to map original text and eliminate the need to store original mappings, saving memory while noting that encrypted text can be up to 30 characters long compared to 10 characters for placeholders (00:35:25). Aarvin George questioned who would manage the overhead of redacting and maintaining local information (00:38:22).



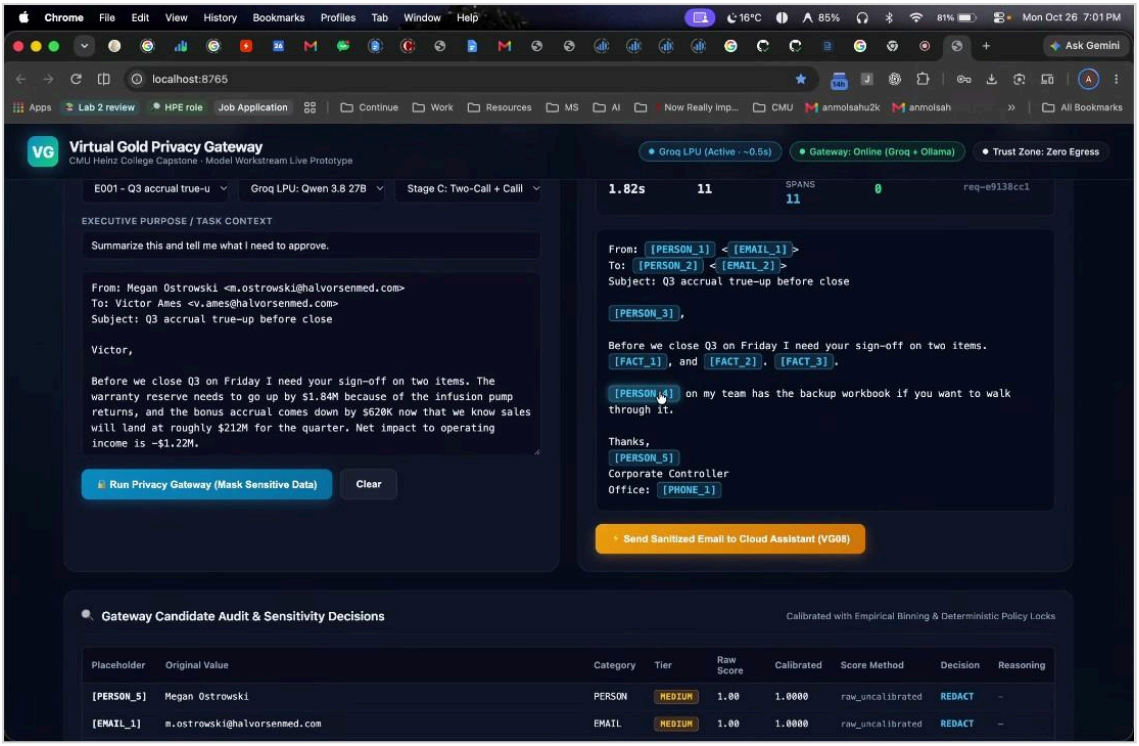
Request Identification and Restoration Design: Anmol Sahu and Mark Chu initially suggested that clients send a specific request ID with API calls (00:38:22) (00:41:11). Aarvin George countered that generating request IDs from the client side did not make sense, proposing instead that the server generate a hash key during redaction and return it with the redacted message so clients can track and return it during restoration (00:42:18) (00:46:31). Inderpal Bhandari confirmed that redaction is exclusively for the primary user (00:43:37). Anmol agreed to this revised design where the input is solely the query, and the server returns the redacted response along with the hash key. Anmol committed to drawing a flow diagram and emailing it to confirm alignment (00:46:31) (00:52:05).

Project Timeline and Integration Next Steps: Inderpal Bhandari asked if the system was ready to hook up to their agent for testing, and Aarvin George confirmed plans to integrate next week on Monday (00:54:40) (00:58:38). Anmol Sahu provided a system demonstration (00:56:02). Mark Chu reviewed the project timeline leading to a final deadline of November 30th, stating they will hold an internal meeting tomorrow to double-confirm the timeline, synchronize on Slack, and address hardware resource requirements from CMU (00:57:28).



Contract Distribution Request: Aarvin George requests that Anmol Sahu send the contract to the connected party (00:59:58).

Email Redaction Pipeline Demonstration: Anmol Sahu demonstrates the email redaction pipeline using an example internal email memo concerning medical devices for quarter 3, highlighting how person names, email addresses, facts, monetary amounts, and other confidential information are successfully redacted across different tiers and categories (00:59:58).



Restoration and Encryption Implementation: Anmol Sahu explains that redacted information can be restored using stored mappings and confirms that encryption and decryption will be utilized moving forward with a Universally Unique Identifier and hash key associated with each conversation and request, replacing placeholders with encrypted hash outputs. Aarvin George clarifies and confirms that one hash key will be linked to each individual conversation and request (01:01:43).

Meeting Adjournment: Mark Chu concludes the meeting, and Mark Chu, Aarvin George, Anmol Sahu, Inderpal Bhandari, and Urte Jesina exchange parting remarks and goodbyes (01:01:43).

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